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//Binary Search Tree

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
struct Node{
 char name[30];
 char mobile[15];
 struct Node *left ,*right;
 };
struct Node*createNode(char name[],char mobile[]) //create new mode
{  struct Node*newNode;
     newNode=(struct Node*)malloc(sizeof(struct Node));
     strcpy(newNode->name,name);
     strcpy(newNode->mobile,mobile);
    newNode->left=NULL;
     newNode->right=NULL;
     return newNode; }
struct Node*insert(struct Node*root,char name[],char mobile[]) // insert
{  if(root==NULL)
     return createNode(name,mobile);
   if(strcmp(name,root->name)<0)
        root->left=insert(root->left,name,mobile);
   else if (strcmp(name,root->name)>0)
        root->right=insert(root->right,name,mobile);
     return root; }
void inorder(struct Node*root)  //inorder
 {  if(root!=NULL)
     { inorder(root->left);
       printf("%s\t%s\n",root->name,root->mobile);
       inorder(root->right);
     }
 }
void preorder(struct Node*root)  // preorder
{ if(root!=NULL)
    { printf("%s\t%s\n",root->name,root->mobile);
        preorder(root->left);
        preorder(root->right);
    }
}
void postorder(struct Node*root) //postorder
{ if(root!=NULL)
     { postorder(root->left);
       postorder(root->right);
       printf("%s\t%s\n",root->name,root->mobile);
    }
}
void search(struct Node*root,char name[])//search
```

```
{ if(root==NULL)
  { printf("Contact not found.\n");
    return;
  }
  if (strcmp(name,root->name)==0)
  {  printf("\nContact Found!\n");
      printf("Name:%s\n",root->name);
      printf("Mobile:%s\n",root->mobile);
  }
  else if(strcmp(name,root->name)<0)
     search(root->left,name);
  else
    search(root->right,name);}
int main()
{ struct Node*root=NULL;
  int choice;
  char name[30],mobile[15];
do
  {  printf("\nONLINE DIRECTORY SYSTEM\n");
      printf("1.create/insert\n");
      printf("2.Display ALL Traversals\n");
      printf("3.Search\n");
      printf("4.Exit\n");
     printf("Enter your choice:");
      scanf("%d",&choice);
      switch(choice)
      { case 1: printf("Enter name:");
          scanf("%s",name);
          printf("Enter mobile number:");
          scanf("%s",mobile);
          root=insert(root,name,mobile);
          printf("Contact added successfully,\n");
          break;
    case 2: printf("\n INORDER TRAVERSAL\n");
        printf("Name\tMobile\n");
        inorder(root);
       printf("\n PREORDER TRAVERSAl \n");
        printf("Name\tMobile\n");
        preorder(root);
       printf("\n POSTORDER TRAVERSAL  \n");
        printf("Name\tMobile\n");
        postorder(root);
 case 3: printf("Enter name to search:");
        scanf("%s",name);
        search(root,name);
        break;
 case  4: printf("Existing progarm\n");
        break;
```

```
    default:
       printf("Invalid choice!\n");
     }
  }while(choice!=4);
  return 0;
}
```

Output:

```
ONLINE DIRECTORY SYSTEM
1.create/insert
2.Display ALL Traversals
3.Search
4.Exit
Enter your choice:1
Enter name:Tanvi
Enter mobile number:1324536
Contact added successfully,
ONLINE DIRECTORY SYSTEM
1.create/insert
2.Display ALL Traversals
3.Search
4.Exit
Enter your choice:1
Enter name:akash
Enter mobile number:3456281
Contact added successfully,
ONLINE DIRECTORY SYSTEM
1.create/insert
2.Display ALL Traversals
3.Search
4.Exit
Enter your choice:1
Enter name:Tanmay
Enter mobile number:46183020
Contact added successfully,
ONLINE DIRECTORY SYSTEM
1.create/insert
2.Display ALL Traversals
3.Search
4.Exit
Enter your choice:2
INORDER TRAVERSAL
Name    Mobile
Tanmay  46183020
Tanvi   1324536
akash   3456281
PREORDER TRAVERSAl
Name    Mobile
```

Tanvi 1324536
Tanmay 46183020
akash 3456281
POSTORDER TRAVERSAL
Name Mobile
Tanmay 46183020
akash 3456281
Tanvi 1324536
Enter your choice:3
Enter name to search:Rutuja
Contact not found.